

Computer configuration and Software installation



SUMMARY

I- MY COMPUTER SPECIFICATIONS

II- THE VIRTUAL ENVIRONMENT OF WORK(ROBOT_PROJECT)

III- LANGUAGES AND SOFTWARE INSTALLATION:

- 1. Python 3.11 for OpenCV**
- 2. Docker and ROS**
- 3. Gazebo**

Introduction

This part of the project is to explain the installation and the configuration of all the software needed for the project; before doing this, I will check my computer specification to know if it can support the software.

I- MY COMPUTER SPECIFICATIONS

1-To check my ram: **free -h**

	total	utilisé	libre	partagé	tamp/cache	disponible
Mem:	14Gi	4,2Gi	6,9Gi	106Mi	3,4Gi	10Gi
Échange:	9,3Gi	0B	9,3Gi			

It means I have a total memory of 14 GB and 6.9 GB is free. The RAM is the memory used by the computer to perform operations, and it's sufficient for this project.

2-To see the GPU: **lspci | grep -i vga**

```
04:00.0 VGA compatible controller: Advanced Micro Devices, Inc. [AMD/ATI] Barcelo (rev c5)
```

It means the GPU is an AMD/ATI Barcelo (Radeon integrated).

3- To see my ubuntu version: **lsb_release -a**

```
No LSB modules are available.
Distributor ID: Ubuntu
Description:    Ubuntu 25.04
Release:        25.04
Codename:       plucky
```

This Ubuntu version is too recent for ROS Noetic used for this project. ROS Noetic need Ubuntu 24.

4-To check the disk space: **df -h /**

```
Sys. de fichiers Taille Utilisé Dispo Uti% Monté sur
/dev/nvme0n1p4    46G   18G   26G  41% /
```

The disk allow us to stock 26 GB of information.

Summary of my computer specifications :

OS	Ubuntu 25.04	Too recent
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RAM	14 GB	Perfect
GPU	AMD Barcela	CPU mode only
Disk	26 GB free	Enough
Drive	NVMe SSD	Very fast

The problem with computer characteristics

1-My computer only have CPU

My computer only has a CPU, which means it can perform one operation at a time, and that is a problem for using Machine Learning for crop recognition. GPUs allow for several operations to be done in parallel, which is why they are useful for Machine Learning.

But the car we want to use is the LIMO PRO, which has the following characteristics:

Processor	NVIDIA Orin Nano
GPU cores	1024 CUDA cores
AI power	40 TOPS
RAM	8 GB LPDDR5
Storage	64 GB eMMC

Solution :

My own computer can't be used for YOLOv5s training because of its lack of a GPU, but I can do the YOLOv5s training directly on the LIMO according to its characteristics.

2-My Ubuntu version is too recent for ROS

Indeed, we need Ubuntu 20.04 for ROS1.

Solution:

We will use Docker and work on it. We can directly work on the LIMO, but we choose to work on Docker to enable us to continue working outside the Lab.

II- THE VIRTUAL ENVIRONMENT OF WORK(ROBOT PROJECT)

Since OpenCv as its own Python version, to avoid a conflict of Python with my original computer Python version I created a virtual environment for executing the OpenCV code.

ENTER the environment

source ~/robot_project/bin/activate

→ (robot_project) appears

EXIT the environment

deactivate

→ (robot_project) disappears

Re-enter anytime

source ~/robot_project/bin/activate

→ (robot_project) appears again

III- LANGUAGES AND SOFTWARE INSTALLATION

1-Python:

We will use OpenCV, YOLOv5s, and ROS for the project, and each one of them needs:

- OpenCV needs: Python 3.13
- YOLOv5s needs: Python 3.11
- ROS needs: Python 3.8

Python version on computer:

Python 3.13.3

Unfortunately, this version only matches with OpenCV.

Solution:

We will install Python 3.11 for YOLOv5s and for ROS we don't need because Python 3.8 Noetic is in Docker.

Python 3.11 installation: `Sudo apt install -y python3.11 python3.11-venv python3.11-dev`

error during installation:

```
Erreur : Impossible de trouver le paquet python3.11
Erreur : Impossible de trouver de paquet correspondant à l'expression rationnelle « python3.11 »
Erreur : Impossible de trouver le paquet python3.11-venv
Erreur : Impossible de trouver de paquet correspondant à l'expression rationnelle « python3.11-venv »
Erreur : Impossible de trouver le paquet python3.11-dev
Erreur : Impossible de trouver de paquet correspondant à l'expression rationnelle « python3.11-dev »
```

This problem comes from the absence of Python 3.11 in the default repository of Ubuntu 25.04 I have, so to continue the installation we need to add it.

Installation by building Python 3.11 from source

This is the most reliable method. Run each command one by one:

1-Remove the broken deadsnakes repository

```
sudo add-apt-repository --remove ppa:deadsnakes/ppa
sudo apt update
```

2- Install build dependencies

```
sudo apt install -y build-essential zlib1g-dev \
libncurses5-dev libgdbm-dev libnss3-dev \
libssl-dev libsqlite3-dev libreadline-dev \
libffi-dev curl libbz2-dev
```

3- Download Python 3.11 source code

```
cd /tmp
curl -O https://www.python.org/ftp/python/3.11.9/Python-3.11.9.tgz
```

4- Extract the source code

```
tar -xzf Python-3.11.9.tgz
cd Python-3.11.9
```

5- Configuration of the build

```
./configure --enable-optimizations
```

6- Compilation of Python 3.11

```
make -j$(nproc)
```

```
kelly@kelly-HP-ProBook-445-14-inch-G10-Notebook-PC:~/XDU_internship.github.io$ python3.11 --version
Python 3.11.9
```

7- Installation of Python 3.11

```
sudo make altinstall
```

8- Verification of the installation

```
python3.11 --version
```

Result:

```
kelly@kelly-HP-ProBook-445-14-inch-G10-Notebook-PC:~/XDU_internship.github.io$ python3.11 -m pip --version
pip 26.1 from /home/kelly/.local/lib/python3.11/site-packages/pip (python 3.11)
```

9- Installation of pip for Python 3.11

```
curl -sS https://bootstrap.pypa.io/get-pip.py | python3.11
```

10-Verification of the pip

python3.11 -m pip --version

Result :

VS CODE EXTENSION

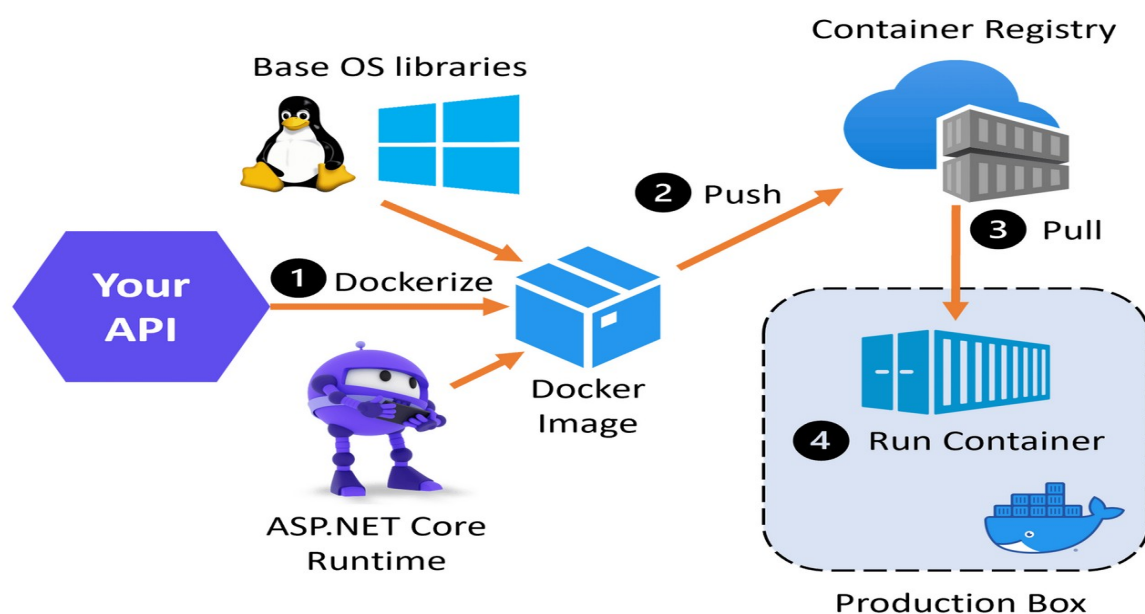
code --install-extension ms-python.python && \:
python extension

code --install-extension ms-vscode-remote.remote-ssh && \:
remote ssh, for accessing LIMO PRO from VS Code.

code --install-extension ms-iot.vscode-ros && \:
To write ROS Code properly.

code --install-extension ms-azuretools.vscode-docker :
For a visual interface of Docker inside of VS Code, to avoid using the commands when we want to perform some operations.

2-Docker and Ros



Since my computer runs ubuntu 25, which isn't compatible with ROS, we chose to run ROS in a Docker container.

Docker installation:

Once Docker is installed, we must restart the computer. If we don't want to restart the computer, we must inform the computer the use Docker by entering "**newgrp docker**" in the terminal.

The first thing we must do every time we turn on my computer is to launch all the container. We do it with the following command:

- docker start -ai ros_kelly

```
kelly@kelly-HP-ProBook-445-14-inch-G10-Notebook-PC:~/XDU_internship.github.io$ docker start -ai ros_kelly
root@46dc513162b0: /#
```

Steps to use ROS on Docker

We need at least three terminals, and to configure them we follow the following commands:

Terminal 1: The master (the bain)

This terminal manages the phonebook so the two other terminals can find each other.

- 1- **Enter the container:** `docker exec -it ros_kelly bash`
- 2- **Sourcing ROS:** `source /opt/ros/noetic/setup.bash`
- 3- **Turn on ROS tools or start the heart:** `roscore`

Terminal 2: The listener (the subscriber)

This terminal represents the robot logic (like a robot arm waiting for a command).

- 1- **enter the container:** `docker exec -it ros_kelly bash`
- 2- **Sourcing ROS:** `source /opt/ros/noetic/setup.bash`
- 3- **Listening to the channel:** `rostopic echo /chatter`

Now, the Terminal 2 is waiting for a command coming from the Terminal 3.

Terminal 3: the talker (the publisher)

This terminal represents the sensors (like the LIMO PRO camera or LiDAR).

- 1- **Enter the container:** `docker exec -it ros_kelly bash`
- 2- **Sourcing ROS:** `source /opt/ros/noetic/setup.bash`
- 3- **Send a message:** `rostopic pub -r 1 /chatter std_msgs/String "data: 'LIMO robot is ready'"`

Once we enter the command 3, the message 'LIMO robot is ready' will be displayed in the 2nd terminal.

```
Terminal 1 : the master
Done checking log file disk usage. Usage is <1GB.
started roslaunch server http://46dc513162b0:45267/
ros_comm version 1.17.4

SUMMARY
=====
PARAMETERS
* /rostdistro: noetic
* /rosversion: 1.17.4

NODES
auto-starting new master
process[master]: started with pid [54]
ROS_MASTER_URI=http://46dc513162b0:11311/

setting /run_id to 1b4201f6-4937-11f1-bf65-000000000000
process[rosout-1]: started with pid [55]
started core service [/rosout]

Terminal 2 : the listener
kelly@kelly-HP-ProBook-445-14-inch-G10-Notebook-PC:~$ docker exec -it ros_kelly bash
root@46dc513162b0:/# source /opt/ros/noetic/setup.bash
root@46dc513162b0:/# rostopic echo /chatter
data: "LIMO robot is ready"
---
data: "LIMO robot is ready"
---
data: "LIMO robot is ready"
---
data: "LIMO robot is ready"
---
data: "LIMO robot is ready"
---

Terminal 3 : the talker
kelly@kelly-HP-ProBook-445-14-inch-G10-Notebook-PC:~$ docker exec -it ros_kelly bash
root@46dc513162b0:/# source /opt/ros/noetic/setup.bash
root@46dc513162b0:/# rostopic pub -r 1 /chatter std_msgs/String "data: 'LIMO robot is ready'"
[1]+  Stopped                  rostopic pub -r 1 /chatter std_msgs/String "data: 'LIMO robot is ready'"
root@46dc513162b0:/#
```

3- Gazebo Installation

commands to install Gazebo:

`apt install ros-noetic-gazebo-ros-pkgs ros-noetic-gazebo-ros-control`

After installing Gazebo, when I tap gazebo in the container nothing happens ; it is because gazebo is a graphical interface and Docker is currently « blind », and has no way to send the video signal on my computer screen. Docker is trying to display an image but has no way to do it. To fix this we will bridge Docker (the box) and the host Ubuntu 25.04. To do it we must follow the following steps :

1- On the Ubuntu 25.05 (out of Docker) :

`xhost +local:docker`

result :

```
kelly@kelly-HP-ProBook-445-14-inch-G10-Notebook-PC:~$ xhost +local:docker
non-network local connections being added to access control list
```

It means that the computer is ready to accept visual drawing from Docker container.

2- We must now indicate to the Docker container where to send the drawing, to do this we must set DISPLAY as environment variable to let the software know it can look at the host screen :

`export DISPLAY=:0`

Now we will try to test the graphical link again :

- Let's install a tiny test app : `apt update && apt install x11-apps -y`

- Run the clock : `xclock`

result :

```
root@46dc513162b0:/# xclock
Error: Can't open display: :0
```

This means that even though the two part (Ubuntu 25.04 and the container) have the configurations needed, the tunnel that link the two has not be created.

Solution :

We will recreate a container but this time with a tunnel to send the drawing toward the host (Ubuntu 25.04) screen.

Steps :

1- First we must stop and remove the old container :

`docker stop ros_kelly`

`docker rm ros_kelly`

2- Grant GPU permissions to docker from the host :

`xhost +local:docker`

2- Launch the new container with GPU support :

```
docker run -it \
  --name ros_kelly \
  --env="DISPLAY=$DISPLAY" \
  --env="QT_X11_NO_MITSHM=1" \
  --volume="/tmp/.X11-unix:/tmp/.X11-unix:rw" \
  osrf/ros:noetic-desktop-full
```

Test :

- In the host : `xhost +local:docker`

- Open a master terminal and configure it like above :

1- **Enter the container:** `docker exec -it ros_kelly bash`

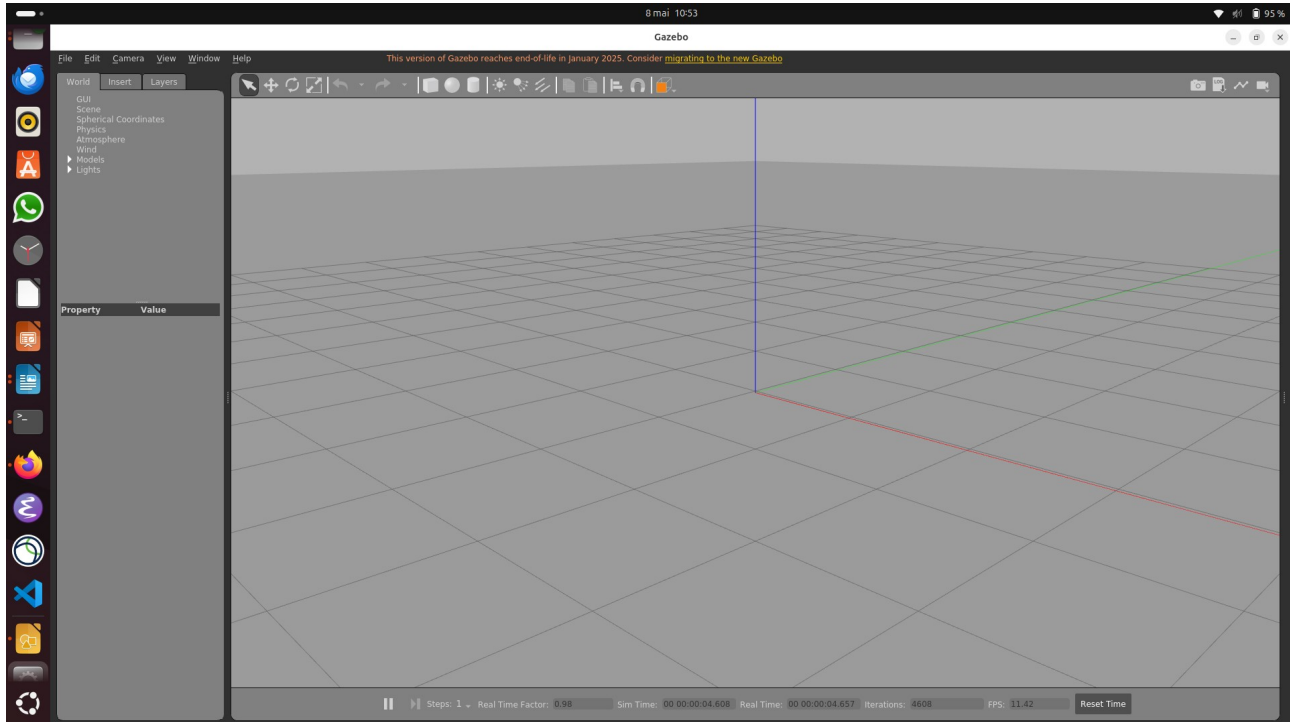
2- **Sourcing ROS:** `source /opt/ros/noetic/setup.bash`

3- **Turn on ROS tools or start the heart:** `roscore`

-Open another terminal (we can call him terminal 4) for send the display :

- 1- **Enter the container:** `docker exec -it ros_kelly bash`
- 2- **Sourcing ROS:** `export DISPLAY=:0`
- 3- **Turn on ROS tools or start the heart:** `gazebo.`

Result :



Error encountered while testing gazebo :

```
roscore http://8b28e5294826:11311/
root@8b28e5294826:/# roscore
... logging to /root/.ros/log/e65f8abc-4ab7-11f1-ba0d-fa383b97ad71/roslaunch-8b2
8e5294826-994.log
Checking log directory for disk usage. This may take a while.
Press Ctrl-C to interrupt
Done checking log file disk usage. Usage is <1GB.

started roslaunch server http://8b28e5294826:43407/
ros_comm version 1.17.4

SUMMARY
=====

PARAMETERS
* /rostdistro: noetic
* /rosversion: 1.17.4

NODES

RLEException: roscore cannot run as another roscore/master is already running.
Please kill other roscore/master processes before relaunching.
The ROS_MASTER_URI is http://8b28e5294826:11311/
The traceback for the exception was written to the log file
```

This means that a master terminal is opened. To close it, we must do :
`pkill -9 roscore; pkill -9 rosout; pkill -9 master.`

TO avoid this error, we can close the master terminal with : `pkill -9 roscore; pkill -9 rosout; pkill -9 master.`

Conclusion for what is done so far

My computer can now run OpenCV, ROS through Docker and Gazebo. My computer has enough capacity to hold the project, except for the YOLOv5s training which will be done directly on the car LIMO PRO.